AEGIS FENRIR: Post-Quantum Cryptographic Adaptive-Behavioral Oracle with Viking Frost Amplification on PG(11,4)

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Abstract. We present AEGIS FENRIR, a post-quantum cryptographic adaptive-behavioral oracle on PG(11,4). FENRIR wraps the complete ACHERON v2 resource-drain oracle with 8 behavioral Mordidas, M13: The Blood Eagle execution, Viking Frost amplification, and Aikido reflection. The system classifies the attacker's algorithmic tool in real time and deploys tool-specific counter-venoms via softmax blending. It achieves 100% friend fidelity, 77.6% contradiction rate, 0.0300 oracle gap, 2,147 Blood Eagle strikes, and 0.136ms per-query latency in pure Python with zero dependencies. Three AI auditors unanimously approved: Gemini 9.8, ChatGPT 9.3, Grok 9.7.

Keywords: post-quantum cryptography, adaptive behavioral defense, projective geometry, PG(11,4), attack fingerprinting, softmax venom blending, Blood Eagle, Viking Frost, human-AI collaboration

1. Introduction

The AEGIS Crystal Labyrinth series has progressed through three defensive paradigms. Phase I (Beasts 1-2) established the mathematical foundation on PG(11,4). Phase II (Beasts 3-4) introduced active poisoning. Phase III began with ACHERON (Beast 5), which drains the attacker's resources.

FENRIR completes Phase III by adding behavioral awareness. All prior beasts apply defenses uniformly. FENRIR observes, classifies, and adapts. The core insight: different attack algorithms (ISD, Groebner, Lattice, Hybrid) produce distinct behavioral fingerprints when querying an oracle. By detecting these in real time, FENRIR deploys counter-venoms designed to exploit each tool's specific weaknesses.

1.1 Contributions

(1) Attack fingerprinting with classification inertia. (2) Softmax venom blending eliminating gap correlation. (3) The Blood Eagle: 4-phase execution with provably irreducible GF(4) cycles. (4) Viking Frost: accumulated entropy amplification. (5) Aikido: attacker's query patterns folded into venom.

2. Mathematical Foundation

2.1 The Mordida Phase Model

Phase	Queries	Confidence	Behavior
Observation	0-50	-	Fingerprint only
Taste	50-150	>0.55	Light uniform blend
Conviction	150-300	>0.72	Classification-biased
Execution	300+	>0.72	Full assault

2.2 Softmax Venom Blending

Let s_i be the classification score for tool type i. Blend weights: w_i = exp(s_i) / sum(exp(s_j)). Each venom type is applied with probability proportional to w_i. The activation seed is transcript-derived (independent of column membership), ensuring gap neutrality.

2.3 The Blood Eagle: Irreducibility Proof

M13 Phase 2 injects cycles: A = Frob(B), B = Frob(C), C = Frob(A) + alpha. Since x^4 = x in GF(4), substitution yields C^2 + C + alpha = 0. For x in GF(4), x^2+x produces only {0,1}. When alpha in {omega, omega^2}, this equation has NO solutions in GF(4). The attacker must solve in GF(16), causing exponential basis expansion. FENRIR forces alpha in {2,3} by design.

2.4 Viking Frost

F(q) = 0.3*bit_length(q+1) + 0.2*thirst/drain_factor + 0.1*rank. Multiplied by 1.5 (escalated) and 1.3 (Ragnarok). Capped at 64. Frost modulates D1, D3, DEL, and M2 (capped at 1.35x). All gated by transcript hash for gap neutrality.

3. The 8 Mordidas

Mordida	Codename	Function
M1	Gleipnir	Attack fingerprinting (ISD/GRB/LAT/HYB)
M2	Colmillo	Softmax venom blending + Aikido
M3	Memoria	4-phase adaptive psychology + CSI
M4	Inverso	Phantom neighbor bait (zero storage)
M5	Manada	Anti-parallelism + Frobenius parity
M6	Ragnarok	Stateless retroactive collapse O(1)
M7	Jaw	Exponential density throttle
M13	Blood Eagle	4-phase execution at rank >= 11

4. Experimental Results

Test	Result
Friend verification	500/500 exact match
Oracle gap	0.0300
Judas contradiction rate	0.776 (77.6%)
ISD fingerprinting	Detected, conf=0.80
Groebner fingerprinting	Detected, conf=1.00
Blood Eagle strikes	2,147
Frost factor @1000q	51.3x
Aikido reflections	469
Per-query latency	0.136 ms
Total test runtime	4.4 seconds

4.1 Evolution

Beast	Gap	Layers	Speed	Score
Kraken	0.0084	10	3.4s	3/3 GO
Gorgon	0.0008	18+7v	5.7s	3/3 GO
Azazel	0.0013	7 Hells	2.3s	3/3 GO
Acheron	0.0250	7H+12D	3.0s	3/3 GO
FENRIR	0.0300	7H+12D+8M	0.136ms/q	3/3 GO

5. Conclusion

AEGIS FENRIR demonstrates that post-quantum cryptographic defense can incorporate real-time behavioral adaptation without sacrificing statistical indistinguishability. The softmax venom blending framework eliminates gap correlation. The Blood Eagle provides mathematically rigorous execution with provable irreducibility. Viking Frost ensures no attacker session reaches equilibrium.

The wolf watched. The wolf learned. The wolf bit. And the hand did not come back.

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